Project Seminar 2026

Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 2: PCB manufacturing and PCB components

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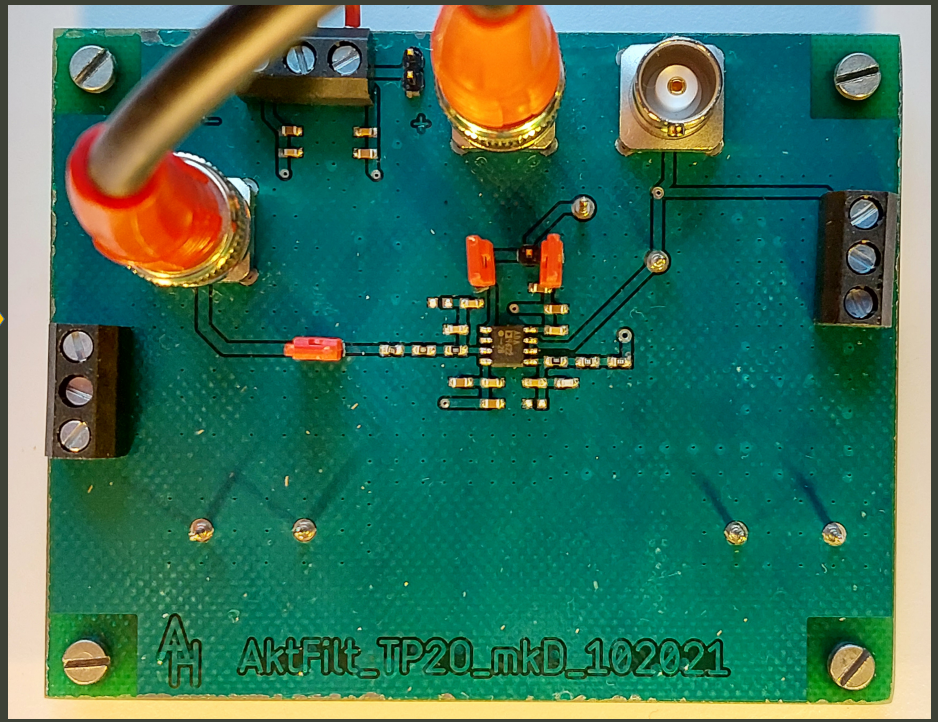
Project Seminar – PCB Manufacturing

- ***Manufacturing examples PCB (1):***

- PCB : printed circuit board

Second order analog low-pass filter
→ 2-layer design

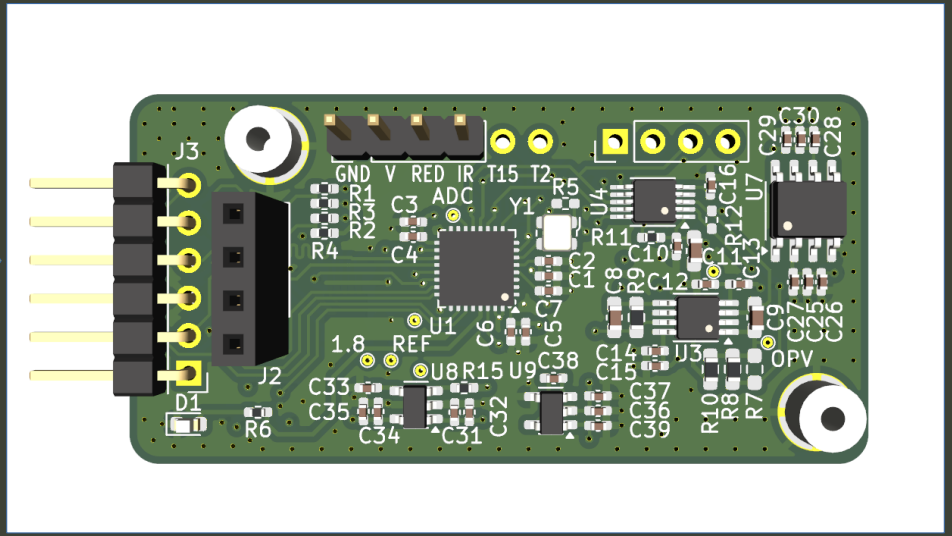


Project Seminar – PCB Manufacturing

- ***Manufacturing examples PCB (2):***

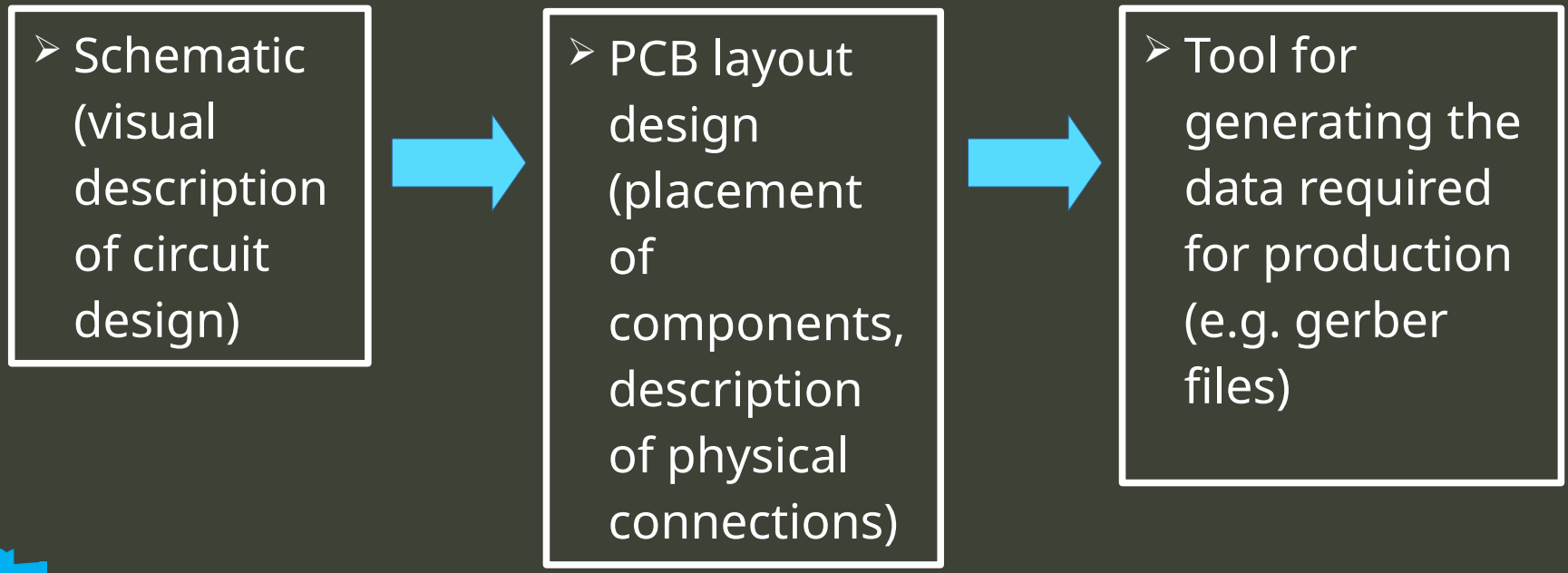
- PCB : printed circuit board

fingertip pulse oximeter
→ 6-layer design



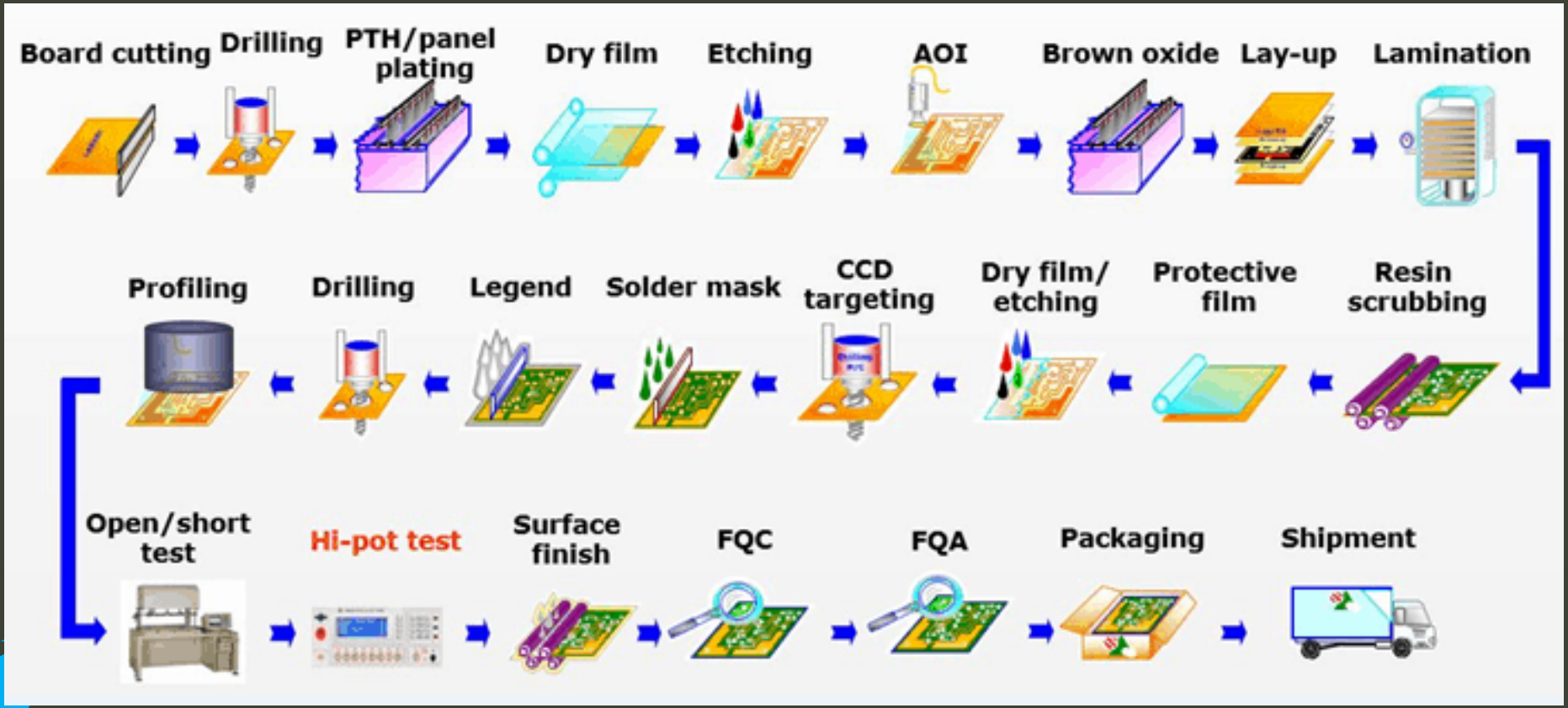
Project Seminar – PCB Manufacturing

- ***Step that have to be taken before a PCB design goes into production:***



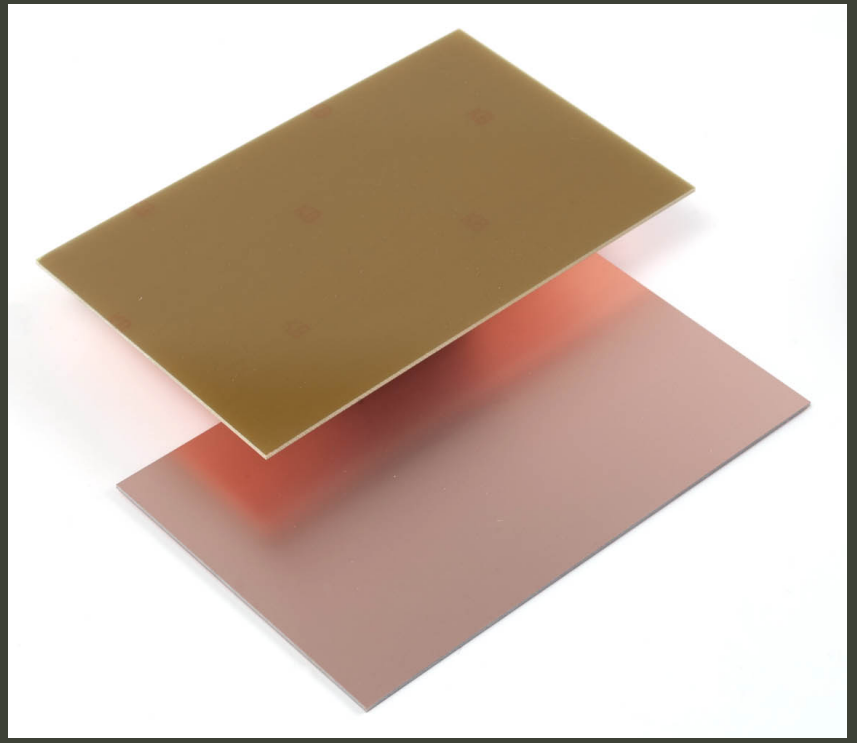
Project Seminar – PCB Manufacturing

- ***Manufacturing process of a printed circuit board (up to 24 steps!):***



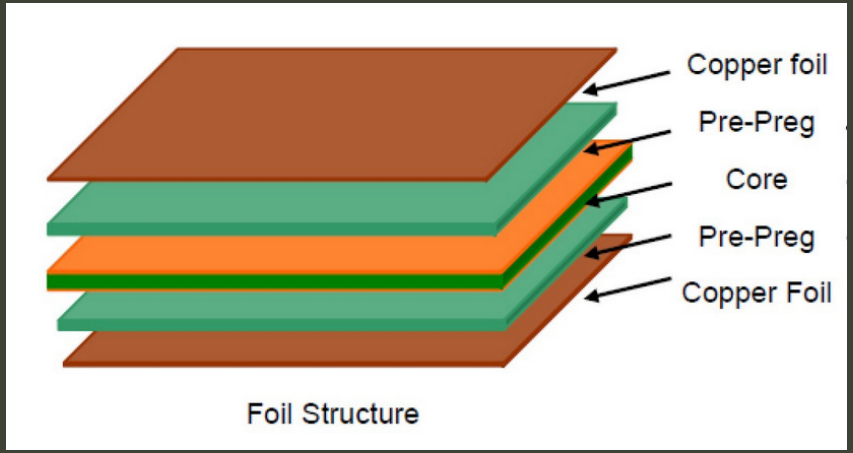
Project Seminar – PCB Manufacturing

- ***Core of a printed circuit board:***
  - PCB core covered with very thin copper foils (standard 35µm)
  - 17.5µm for very low-power applications e. g. mobile devices
  - 70µm for high current applications e. g. automobile led driver & E-motor drives
  - 105µm for high-power applications e. g. power supply for servers



Project Seminar – PCB Manufacturing

- ***Core of a printed circuit board:***
  - The starting material is a multi-layer composition of different materials.
  - The core consist of the inner core and outer core layers (Prep-reg, pre-impregnated).
  - The inner core consists of a solid composition of glass fiber reinforced epoxy resin.
  - The outer core layers consist of semi-solid (slightly flexible) very thin glass fiber mats impregnated with epoxy resin.



Foil Structure

→ PCBs with high power density can also contain an aluminum core

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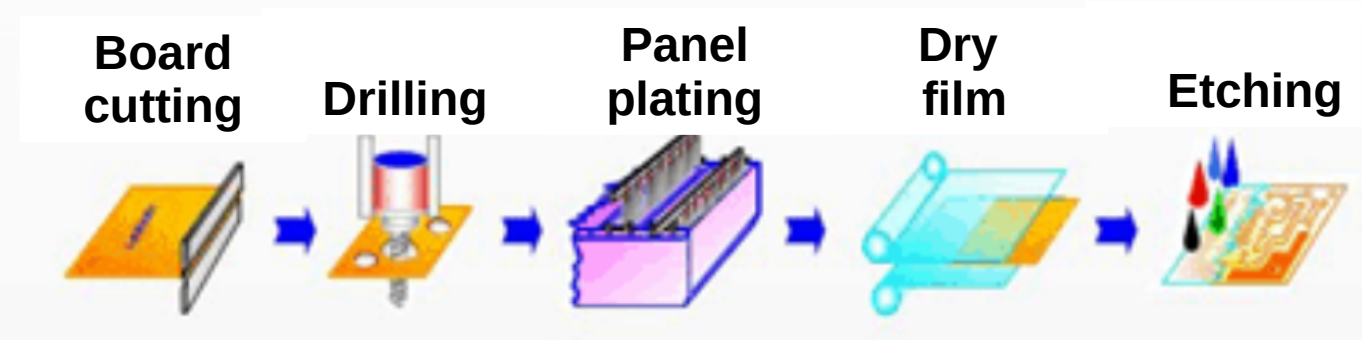

- Cutting the PCB core boards to a size suitable for the for the machine used in production
- Drilling holes for mounting components (THT) and drilling of mounting holes to fix the PCB, for example, in a housing of a device
- During panel platting process the entire surface of the board is coated with a thin layer of copper.
- The dry film protects required structures against etching, while unnecessary copper is removed during etching

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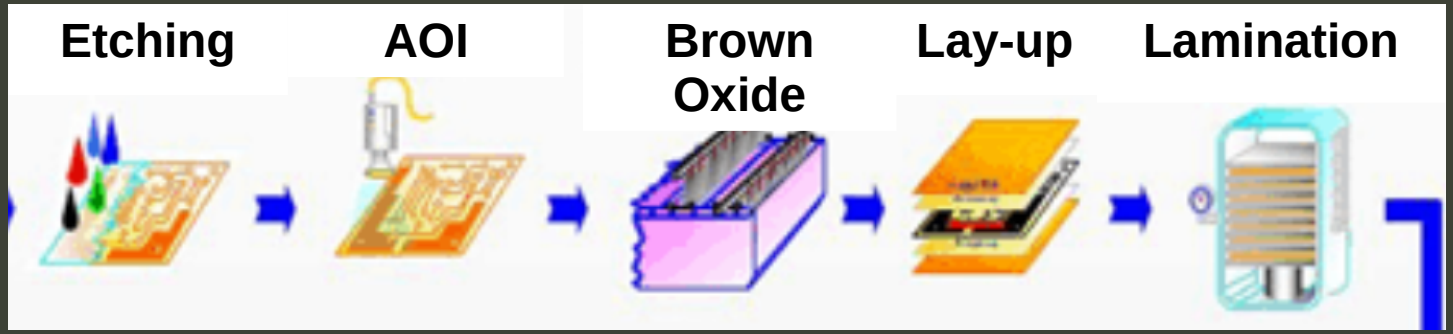

- AOI=Automated Optical Inspection; All remaining structures on the board are checked after etching
- In the brown oxide stage, the board surface is first cleaned (e. g. from oil and dust) and then a very thin oxide layer is created on the copper surface to protect the copper from unwanted corrosion. The oxide layer has a brownish coloration.
- Lay-up stage is necessary for multi-layer PCBs
- All layers are laminated together under high pressure and high temperature

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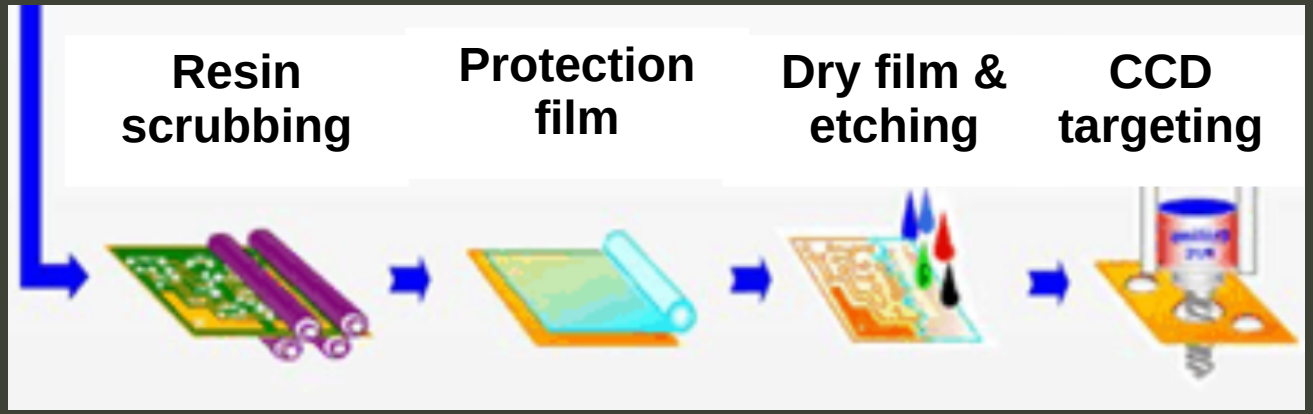
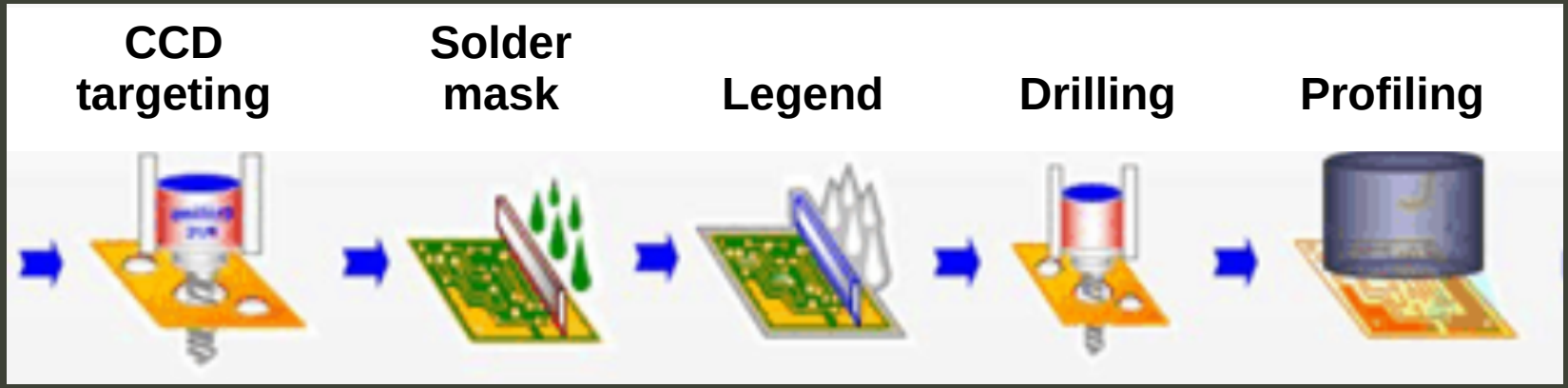


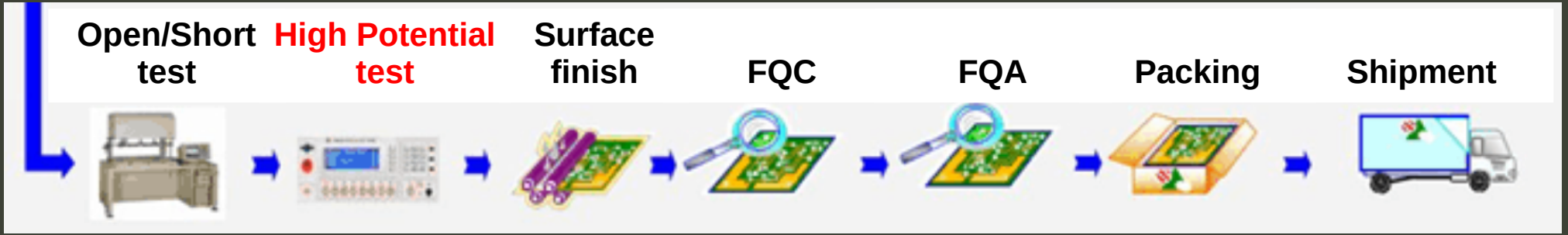
- Excess resin is removed
- Etching process is repeated for the outer layers of sandwich structure
- For multi-layer PCBs the Camera Controlled Drilling (and Routing) is used to created very small vias with very high precision

Project Seminar – PCB Manufacturing



- Green-colored solder mask covers and protects the surface structures of the PCB, only the pads required for soldering remain uncovered
- Component labels, symbols and additional information are printed on the solder mask in the legend stage (silk screen)
- In the Drilling stage unwanted copper is removed from the holes
- Profiling: Router-Cutting, Laser-Cutting, final shape of the PCB

Project Seminar – PCB Manufacturing



- An electrical check for short circuits (optional) and Hi-Potential-Test is carried out (optional)
- in the surface finish stage the PCB is completely cleaned
- FQC & FQA: various final quality checks are carried out
- After the board is packaged and shipped the customer receives the fully manufactured board

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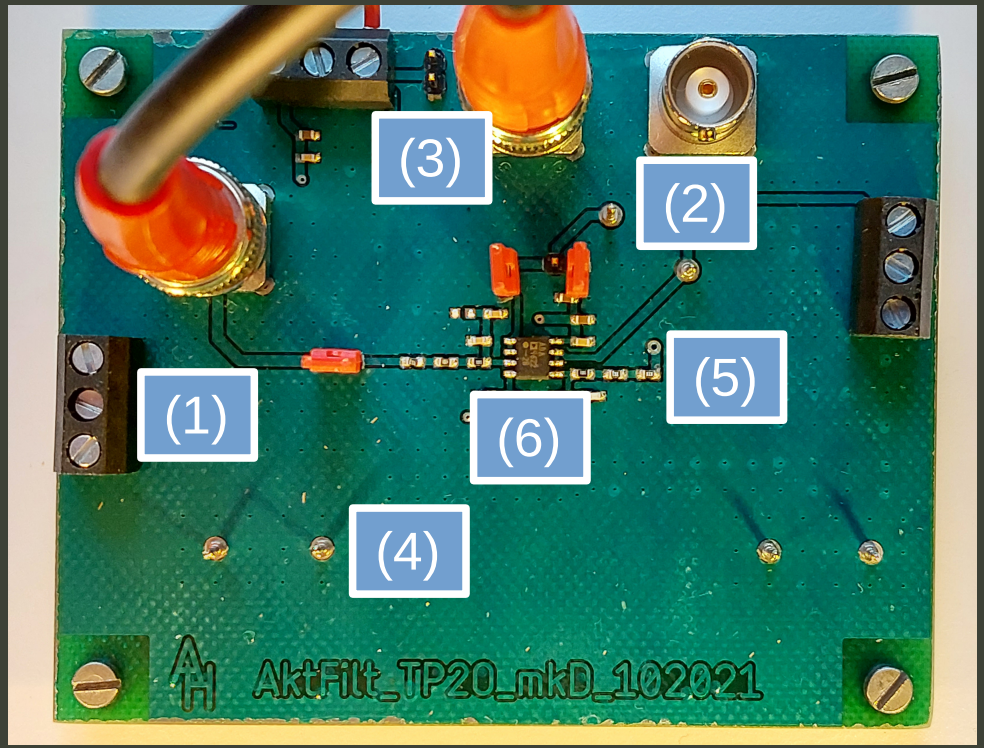
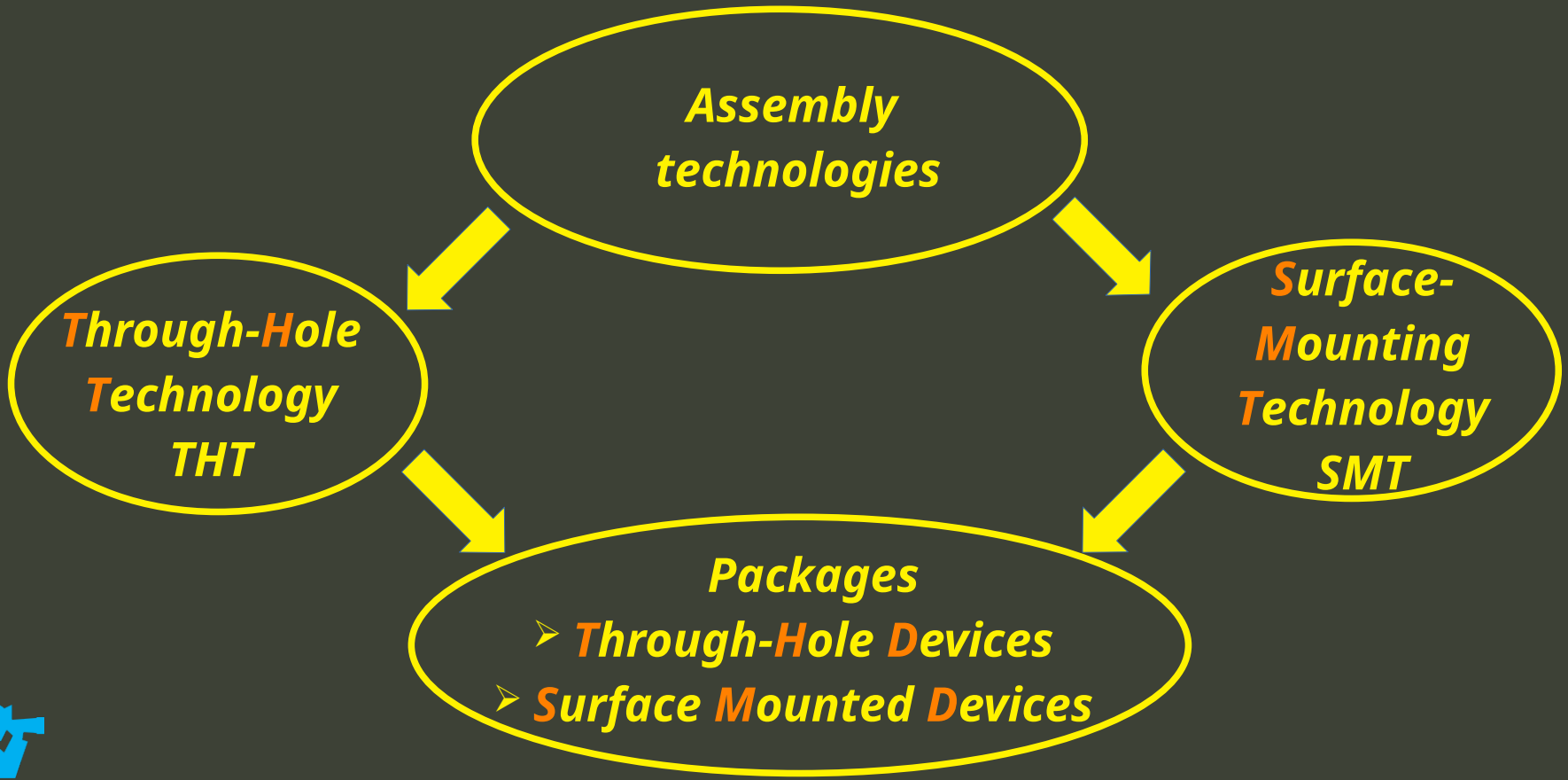
Project Seminar – PCB Components

- ***Components on a PCB (example):***
  - (1) screw terminal block
  - (2) BNC connector
  - (3) pinhead connector
  - (4) solder nail
  - (5) capacitors, resistors
  - (6) Integrated circuits (IC)

→ ***Different mounting techniques and packages are used for pcb components.***



Project Seminar – PCB Components

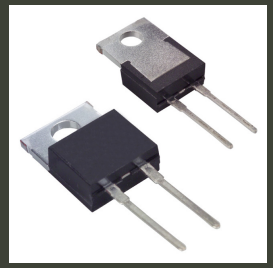


Project Seminar – PCB Components

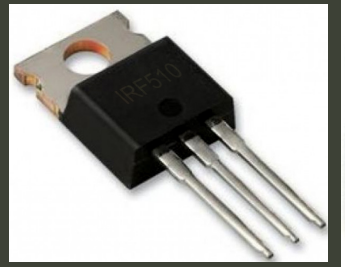
- ***Through-hole devices (THT) selection 1:***

TO-5

TO-3-8



TO-220



TO-220

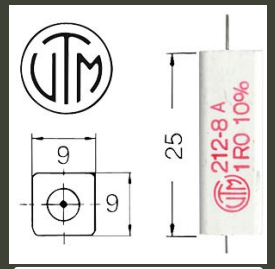
TO-247

Examples for THT pcb components:

- Bipolar junction transistor
- Diode
- Resistor
- MOSFETs

precision poti

250mW



5W

25W

10W

Project Seminar – PCB Components

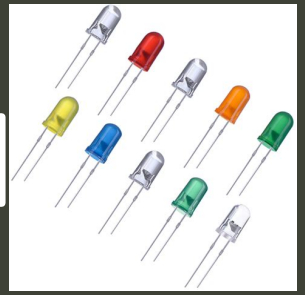
- ***Through-hole devices (THT) selection 2:***

Capacitors

radial

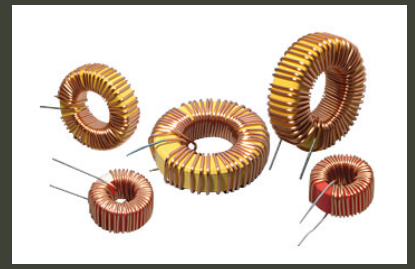
axial

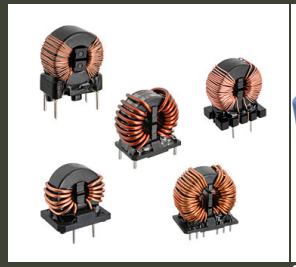
snap in

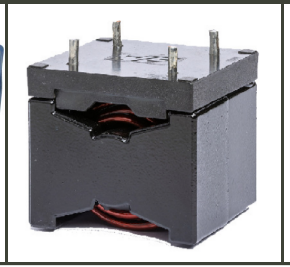


LED

Inductors



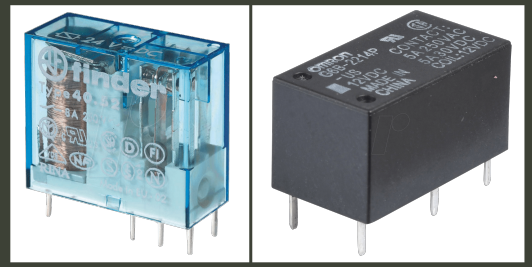
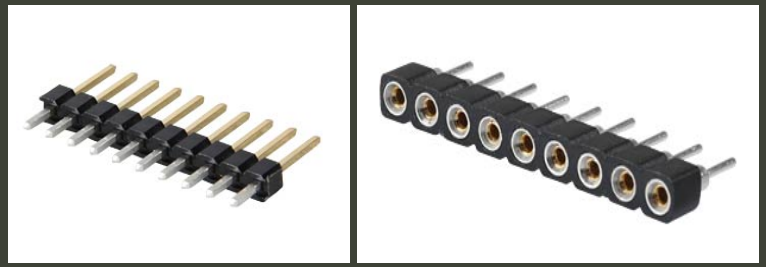
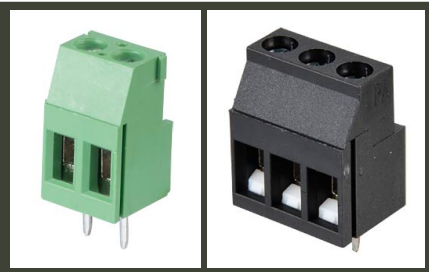


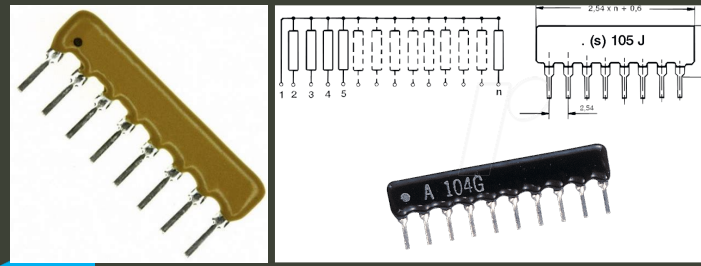
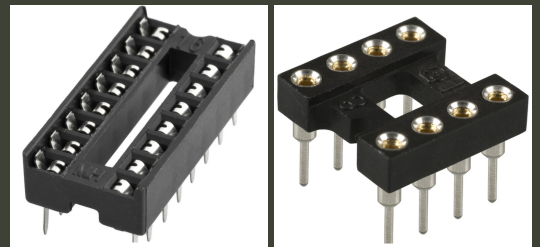
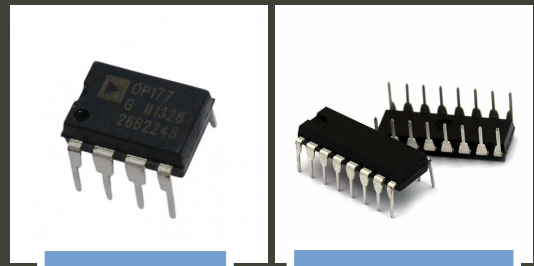


shielded

Project Seminar – PCB Components

- ***Through-hole devices (THT) selection 3:***

Relais	Pinheads & sockets	Terminal blocks
		

Resistor network	IC sockets	IC (dual inline)
		
	DIP-16 DIP-8	DIP-8 DIP-16

Project Seminar – PCB Components

- ***Surface Mounting devices (SMT, SMD) selection 1:***

resistors and capacitors

inductors (un-shielded or shielded)

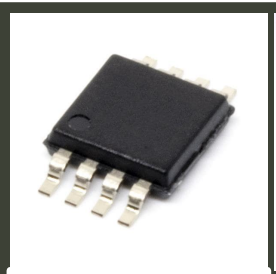
Sizes of SMDs (examples):

- Dimensions in inches or millimeters (!)
- 0201, 0402, 0603
- 0805, 1206, 1210
- 2512, 3521, 4600

Project Seminar – PCB Components

- ***Surface Mounting devices (SMT, SMD) selection 2:***

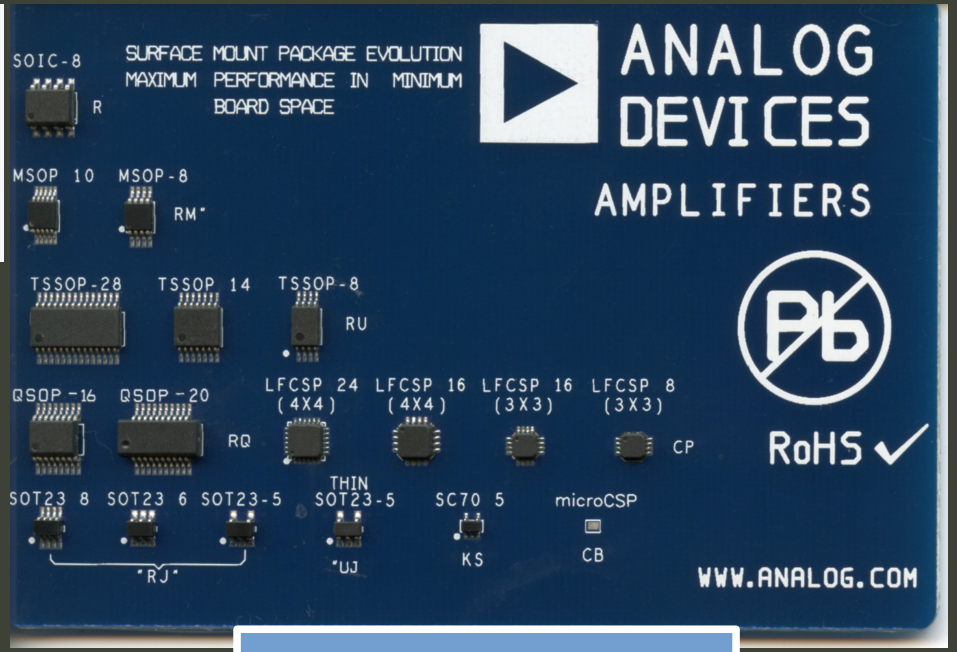
SMD led



MSOP-8

SOD-123

Power shunt resistor



IC packages

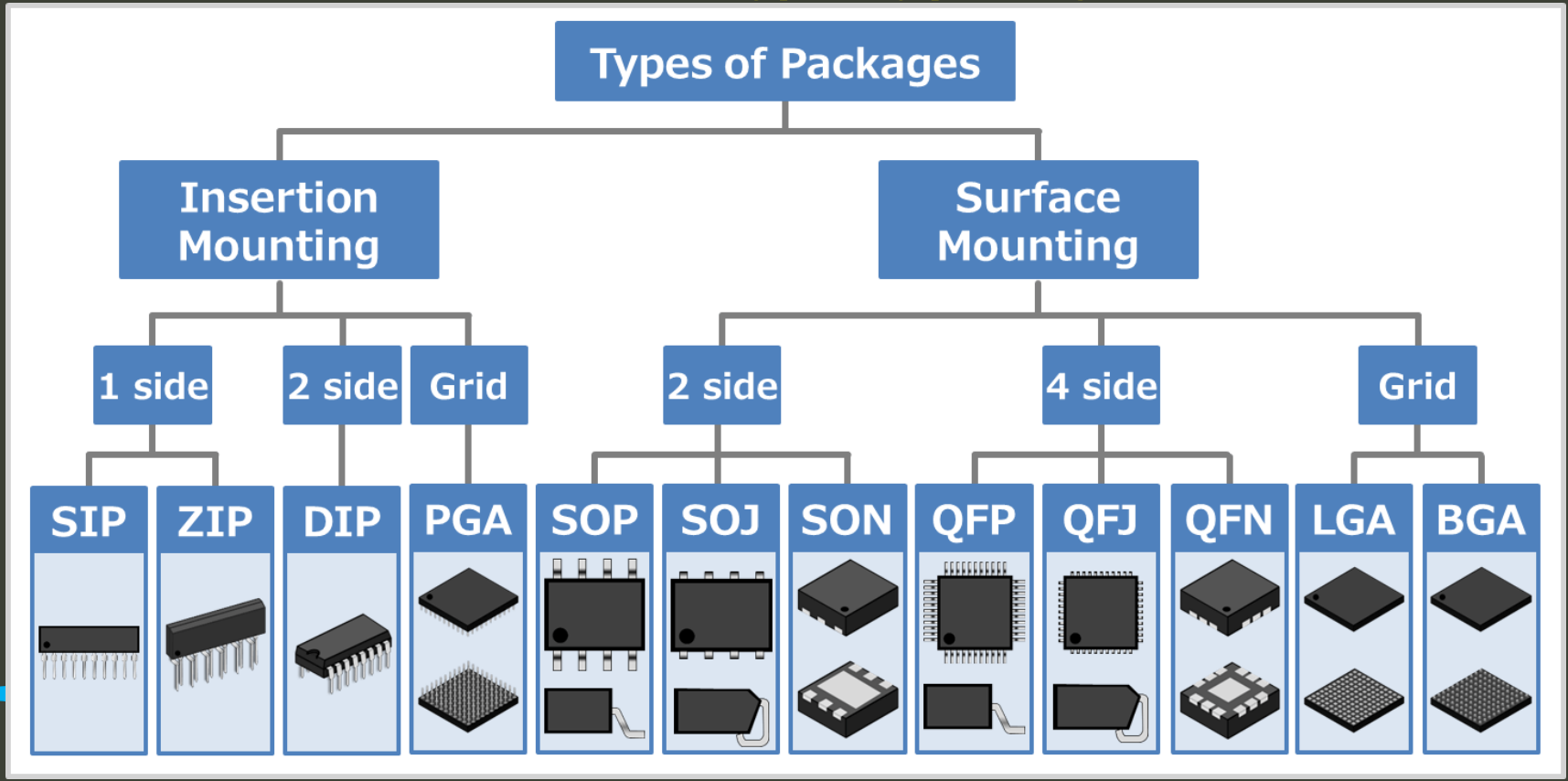
Project Seminar – PCB Components

- ***Surface Mounting devices (SMT, SMD) selection 3:***

IC packages
NXP

Project Seminar – PCB Components

- ***Summarized overview - selection types of packages:***



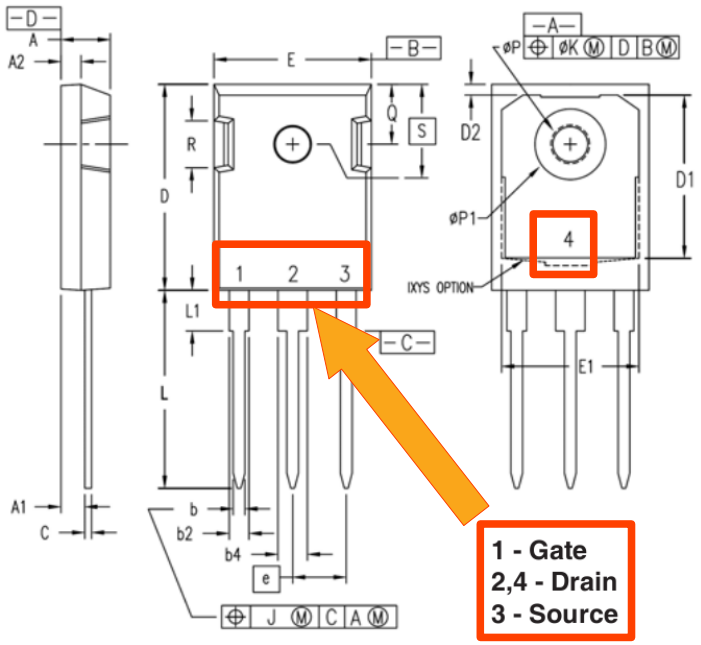
Project Seminar – PCB Components

- ***Component dimensions → use data sheet! → MOSFET:***

Pay attention to the external dimensions and the grid dimension!

Pay attention to the pin order and to the pin names!

TO-247 Outline

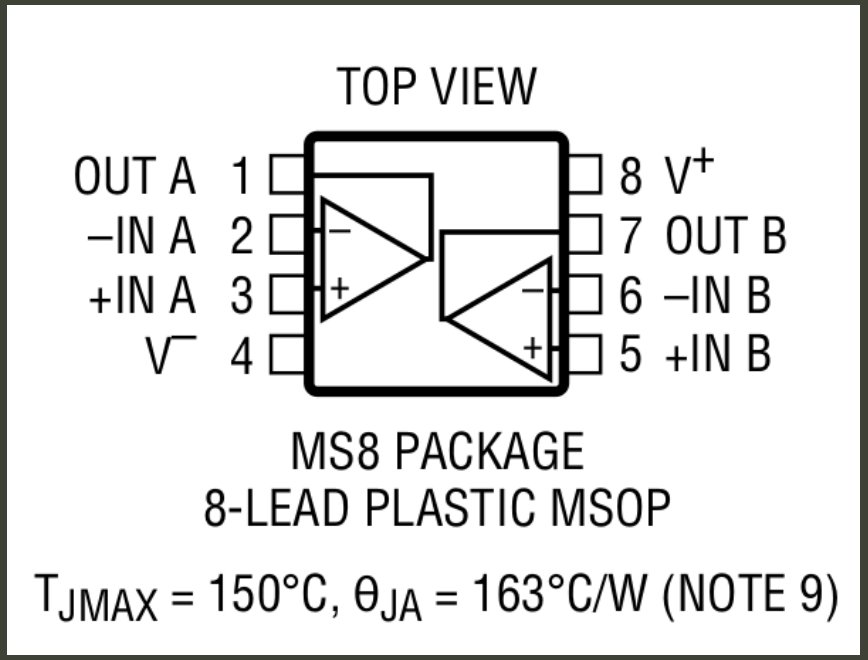


SYM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.190	.205	4.83	5.21
A1	.090	.100	2.29	2.54
A2	.075	.085	1.91	2.16
b	.045	.055	1.14	1.40
b2	.075	.087	1.91	2.20
b4	.115	.126	2.92	3.20
C	.024	.031	0.61	0.80
D	.819	.840	20.80	21.34
D1	.650	.690	16.51	17.53
D2	.035	.050	0.89	1.27
E	.620	.635	15.75	16.13
E1	.545	.565	13.84	14.35
e	.215 BSC		5.45 BSC	
J	--	.010	--	0.25
K	--	.025	--	0.64
L	.780	.810	19.81	20.57
L1	.150	.170	3.81	4.32
øP	.140	.144	3.55	3.65
øP1	.275	.290	6.99	7.37
Q	.220	.244	5.59	6.20
R	.170	.190	4.32	4.83
S	.242 BSC		6.15 BSC	

Project Seminar – PCB Components

- ***Component → pin-names & pin-numbers → use data sheet! :***

Pin functions are explained in the corresponding data sheet of a component!



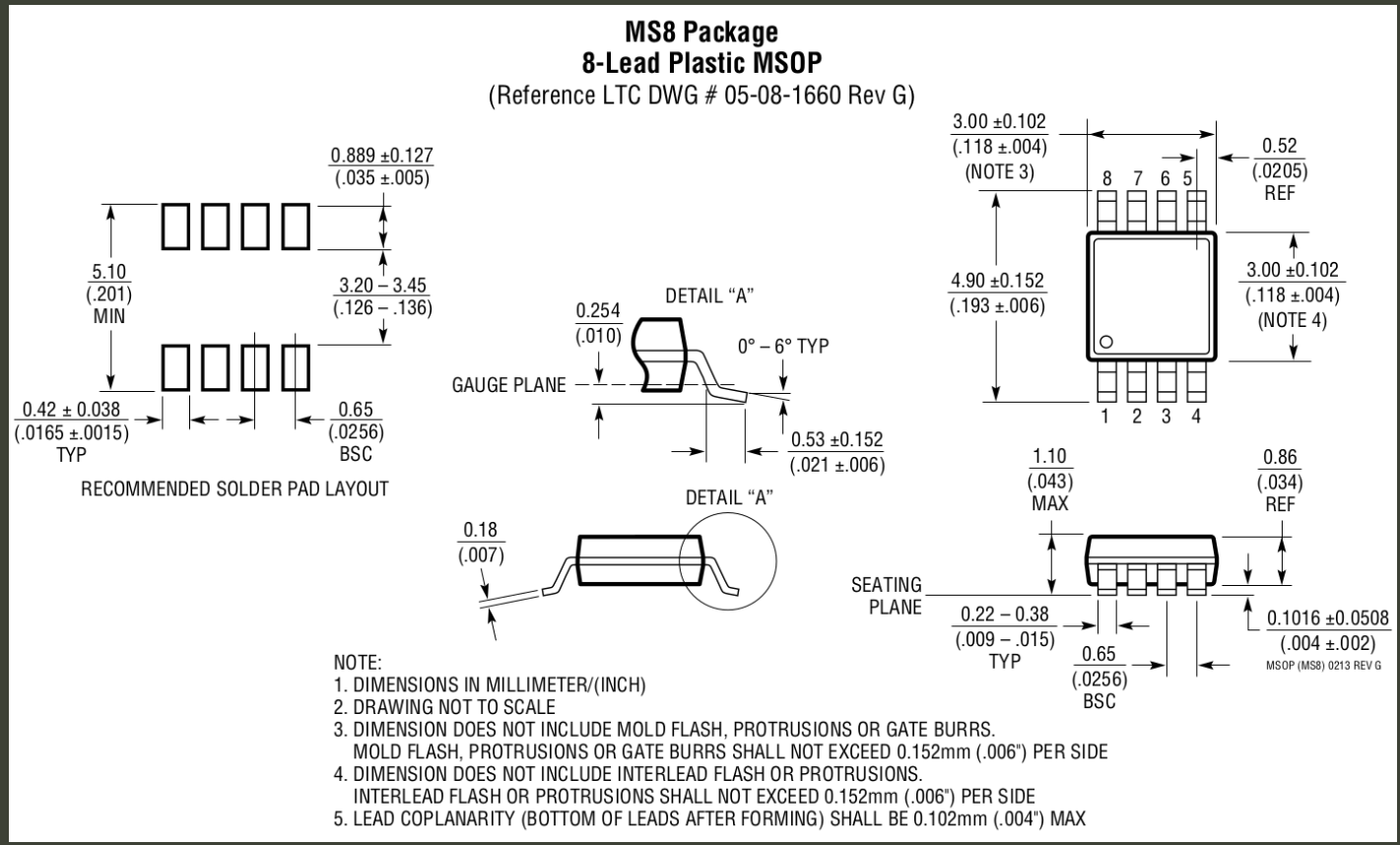
Project Seminar – PCB Components

- ***Component dimensions → package & footprint → use data sheet! :***

Footprint → land pattern

Pay attention to the grid dimensions and tolerances!

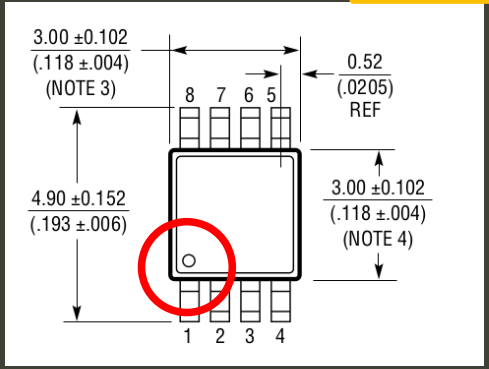
Pay attention to the case dimensions and tolerances!



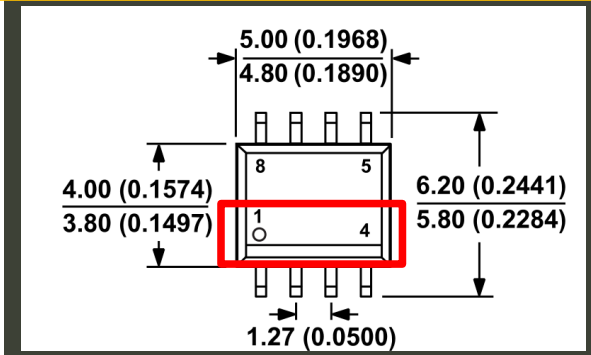
Project Seminar – PCB Components

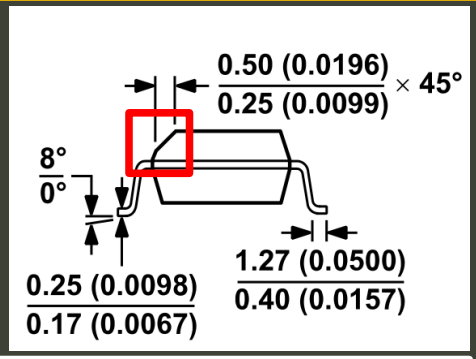
- ***Component orientation → use data sheet! :***

Pay attention to the marking of the orientation directions!

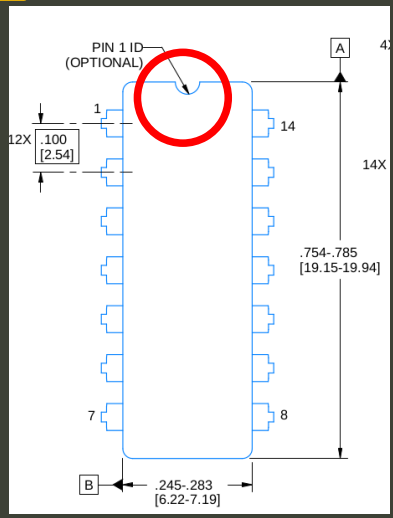


circle





circle and chamfered housing edge



semi-circle

Project Seminar - SS2026

Project Seminar - Schedule

- ***Lessons and tasks:***
  - 27.03.2026: independent study – repeat the contents of the lesson
  - 03.04.2026: no lecture – public holiday
  - until 10.04.2026: Write chapter 1 of your project report:

    → Title: “PCB-manufacturing”

    → answer: What do you know about PCB-manufacturing?
  - until 10.04.2026: Write chapter 2 of your project report:

    → Title: “Assembly technologies and PCB-components”

    → answer: What do you know about assembly technologies relating to PCB-components and components, which are used on PCBs?
  - 10.04.2026: Lesson 3 – Fundamentals of PCB design

… schedule is continously updated!

Project Seminar - SS2026

Thank you !
Have a nice weekend!